Intensity, decibels and Doppler

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish intensity change from frequency shift.

SUCCESS CRITERIA - I can...

1. Use an intensity ratio separately from pitch.
2. Interpret a logarithmic sound-level difference.
3. Predict approach and recession Doppler trends.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **500 Hz** tone lasts **0.3 s**. **Find number of cycles** and period.

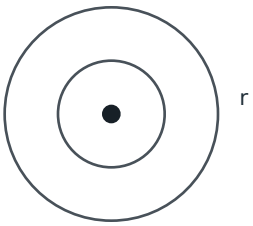
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta\beta = \mathbf{10}\ \log_{10}(I_2/I_1)$.
I is intensity; β is sound level in dB. Tenfold intensity gives **+10 dB**. Approach generally raises observed frequency; recession lowers it.

Worked reasoning: For the same point source, doubling distance spreads its power over four times the area, so ideal intensity drops to one quarter. This assumes spherical spreading without extra absorption.

Your turn: Draw two concentric wavefronts and indicate why the outer surface shares the power more widely.



Label distances and compare intensity.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A moving source approaches a stationary listener, then recedes. **Sketch closer wavefront spacing** in front and wider spacing behind; **compare heard pitch**.

Source moves right. Add wavefronts.



Marzano check - circle: ① ② ③

04 Apply the physics

A sound meter records an intensity increase by a factor of **4**. Use $\Delta\beta = \mathbf{10}\ \log_{10}(I_2/I_1)$ to **find the level change**. Then **compare the intensities of 60 dB** and **80 dB** sounds. **Complete the ratio table**.

Case	Intensity ratio	Level change (dB)
A	____	____
B	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

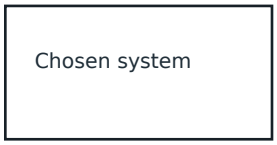
A loud stationary siren and a quiet approaching siren are compared. Does “louder” alone mean “higher frequency”? Explain **0**.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** sample with c=**500 J**/(kg·°C) warms **6°C**. **Find absorbed energy**. If it is **5%** of source energy, find source energy.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Stationary listener; source approaches at **20 m/s**; emits **500 Hz**; v_{sound}=**340 m/s**.
1. Use $f_{obs}=f_s\ v/(v-v_s)$.
2. **Find observed frequency**.

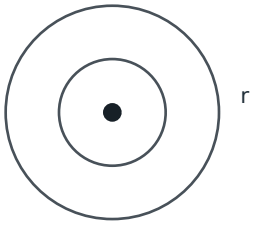
Source moves right. Add wavefronts.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Using $\Delta\beta=$**10** $\log_{10}(I_2/I_1)$, find the level change when distance triples from an ideal point source. Separately explain the higher observed pitch of an approaching source.



Label distances and compare intensity.

Marzano check - circle: 1 2 3